2.35 Show each of following, for all formulas ϕ, ψ, θ.
(a) $\phi \equiv \phi$
(b) If $\phi \equiv \psi$ then $\psi \equiv \phi$.
(c) If $\phi \equiv \psi$ and $\psi \equiv \theta$, then $\phi \equiv \theta$.

Proof of (a). Let v be any truth assignment. Then $v(\phi) = v(\phi)$ so $\phi \equiv \phi$. □

Proof of (b). Suppose $\phi \equiv \psi$, by definition we then have $v(\phi) = v(\psi)$ for all truth assignments v. So then it is also true that $v(\psi) = v(\phi)$ for all truth assignments v. Thus, $\psi \equiv \phi$. □

Proof of (c). Suppose $\phi \equiv \psi$ and $\psi \equiv \theta$. By definition, $v(\phi) = v(\psi)$ and $v(\psi) = v(\theta)$ for all truth assignments v. So then it is also true that $v(\phi) = v(\theta)$ for all truth assignments v and so $\phi \equiv \theta$. □

2.34 Show that $\phi \equiv \psi$ if and only if $(\phi \iff \psi)$ is a tautology, for all formulas ϕ, ψ.

Proof. ($\Rightarrow$): Suppose $\phi \equiv \psi$. By definition for all truth assignments, v, $v(\phi) = v(\psi)$. Let v be any truth assignment. Then $v(\phi) = v(\psi)$, so by the truth table for $\iff$ $v((\phi \iff \psi)) = T$. Since v was arbitrary, $(\phi \iff \psi)$ is a tautology.

($\Leftarrow$): Suppose $(\phi \iff \psi)$ is a tautology. Then for any truth assignment v, we have $v((\phi \iff \psi)) = T$. The truth table shows $\iff$ takes value T only when both sides agree. Hence $v(\phi) = v(\psi)$. Since v was arbitrary, $\phi \equiv \psi$. □